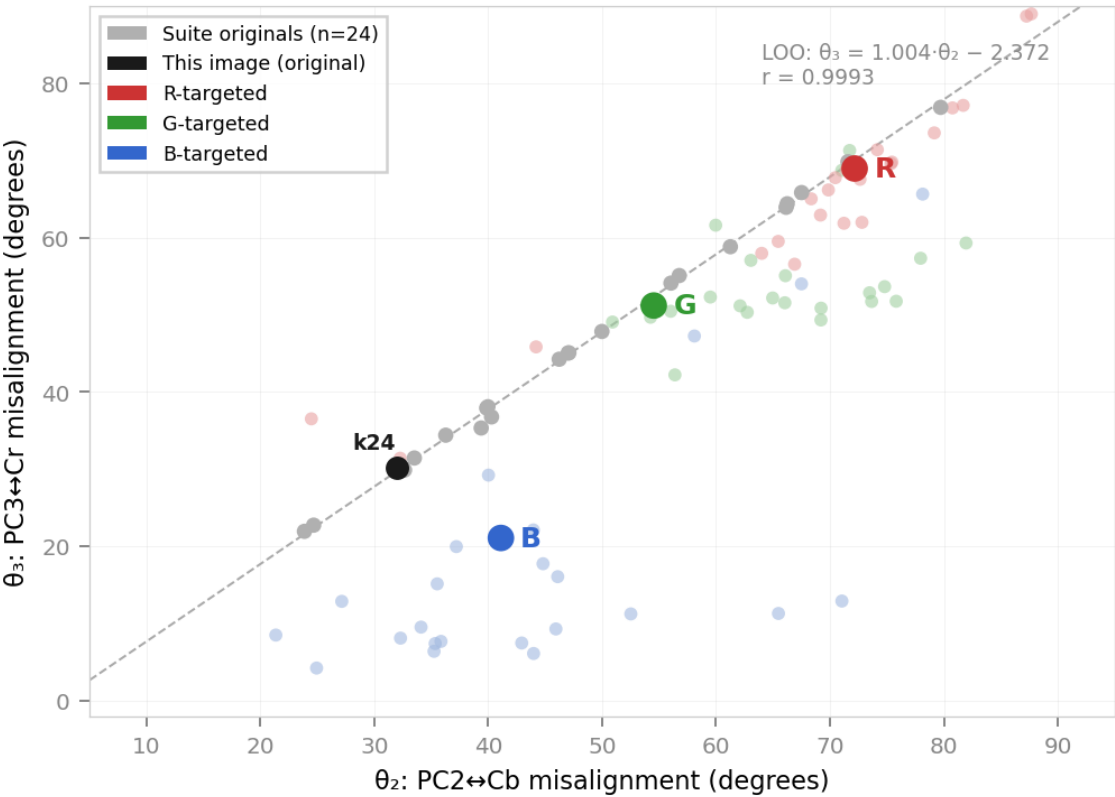


Compression Profile: KODIM24768x512 | 24-bit
RGB | PNG (lossless)
mountain chalet | Near-Degenerate (PC1 = 97.36%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9761	0.9389	0.9649	0.960	32.03°	30.12°	+0.33°
R-targeted	0.2519	0.2600	0.9935	0.502	72.17°	68.99°	-1.10°
G-targeted	0.2515	0.9919	0.2584	0.501	54.53°	51.22°	-1.16°
B-targeted	0.9666	0.1378	0.1670	0.424	41.11°	21.08°	-17.82°

R-targeted isolates red (R-G, R-B < 0.26) while preserving G-B coupling at 0.994.
G-targeted isolates green (R-G, G-B < 0.26) while preserving R-B at 0.992.
B-targeted preserves R-G at 0.967 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold (r = 0.9993).

Compression Profile: KODIM24 768x512 | 24-bit RGB | PNG (lossless)

mountain chalet | Near-Degenerate (PC1 = 97.36%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

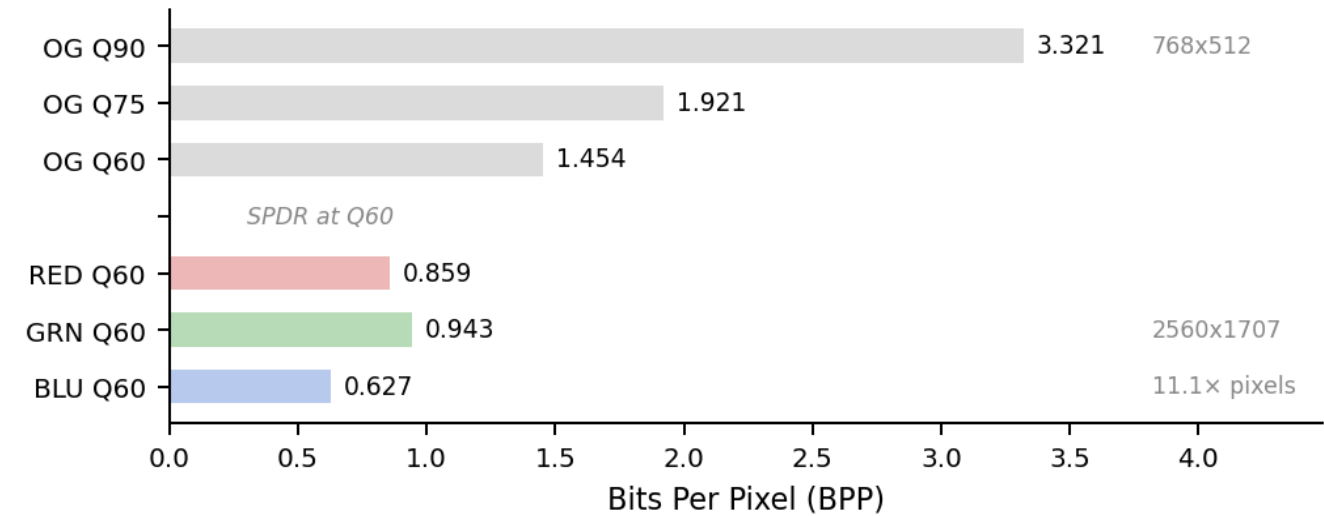
ORIGINAL (kodim24)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	689.8 KB	14.372	393,216
JPEG Q90	JPEG	768x512	159.4 KB	3.321	393,216
JPEG Q75	JPEG	768x512	92.2 KB	1.921	393,216
JPEG Q60	JPEG	768x512	69.8 KB	1.454	393,216

REDISTRIBUTED (kodim24 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	458.4 KB	0.859	4,369,920
GRN Q60	JPEG	2560x1707	502.8 KB	0.943	4,369,920
BLU Q60	JPEG	2560x1707	334.6 KB	0.627	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM24768x512 | 24-bit
RGB | PNG (lossless)

mountain chalet | Near-Degenerate (PC1 = 97.36%)

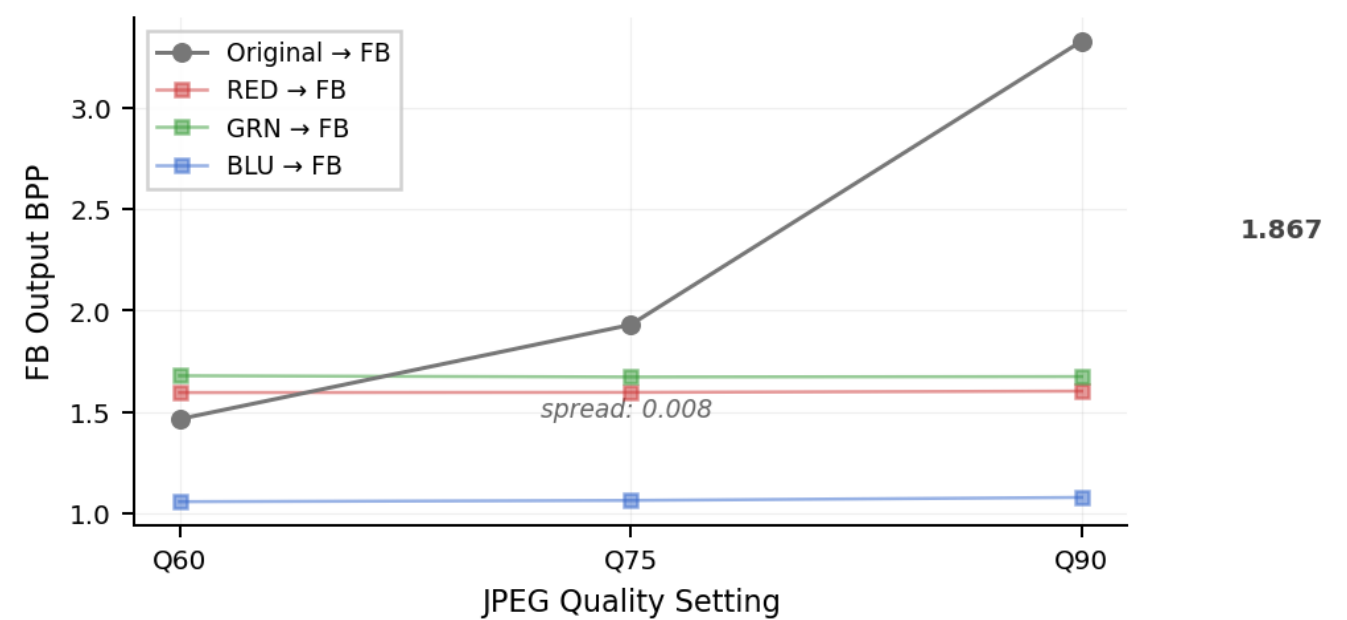
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	3.031	–
R-targeted → FB	2560x1707	2048x1366	1.602	47.1%
G-targeted → FB	2560x1707	2048x1366	1.674	44.8%
B-targeted → FB	2560x1707	2048x1366	1.077	64.5%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.420	53.2%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9618 vs 0.960 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.867 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.008 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim24

Best BPP (Q60, direct)	0.627	2560x1707, 4,369,920 px
Best BPP through FB	1.077	2048x1366 output
vs OG Q90 → FB	67.7%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216